

Specification and verification of systems (IT-449)

Course placement: It is a technical elective as well as MnC elective.

Course format: 3-0-0-3

Course contents: Modelling hardware and software systems through transition systems; modelling concurrency and interleaving in systems; linear time properties; safety properties; liveness properties; Buchi automata, omega regular properties; syntax and semantics of LTL (linear temporal logic); syntax and semantics of CTL (computation tree logic); introduction to model checking tool SPIN.

Selected readings: (1) Principles of model-checking, MIT Press, Joost Pieter Katoen, Christel Baier. (2) Logic in computer sciences, modelling and reasoning about systems, Cambridge press, Michael Huth, Mark Ryan.

Course outcomes: Designing a software or a hardware system is a nontrivial task. The design of the system is formally modelled and then verified to gain confidence. Model checking is one of the predominant techniques for verification of systems. This course aims to introduce students about the mathematical and logical foundations of model checking.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X								X

Lecture schedule:

Motivation for verification—1 lecture

Modelling software---3 lectures

Modelling hardware---3 lectures

Modelling concurrency—3 lectures

Linear time properties --- 4 lectures

Safety properties ---4 lectures

Liveness properties--- 4 lectures

Buchi automata--- 4 lectures

Omega regular properties--- 3 lectures

LTL syntax and semantics --- 3 lectures

CTL syntax and semantics —3 lectures

Spin model checker --- 5 lectures.

Grading policy: The grade earned by a student will depend on (1) the total marks earned by him in all the exams conducted over the entire semester, (2) the SPIN project, and (3) active participation in the class discussions. The weightage of each part is as follows:

Exam marks: 70 %

Project: 20 %

Class participation: 10 %